

Sean Luka Girgis

Senior Data Engineer | AWS, SQL, Python, Parquet & Production Data Systems

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Professional Summary

Senior Data Engineer with 20+ years of enterprise technology experience across AWS data platforms, SQL, Python, PySpark/Spark, ETL/ELT pipelines, Apache Parquet, production data systems, data quality, documentation, mentoring, and technical standards. Strong background building scalable data workflows, optimizing SQL and reporting layers, troubleshooting production systems, documenting operational procedures, and delivering reliable analytics datasets for business and technical stakeholders. Best aligned to senior data engineering roles requiring AWS, SQL/T-SQL/Postgres-adjacent database depth, Python, Parquet, scalable system design, production support, documentation, mentoring, and cross-functional technical leadership; OpenSearch/Elasticsearch and C# are positioned as adjacent or secondary areas rather than overstated production ownership.

Professional Experience

Senior Capacity & Data Engineer at CITI (2017-11 – 2025-12)

- Built Python/Pandas/PySpark ETL pipelines ingesting telemetry from 6,000+ infrastructure endpoints and transforming raw operational data into curated datasets for reporting, analytics, forecasting, and validation workflows.
- Designed AWS-based data platform components using S3 landing zones, AWS Glue, Athena, Redshift, Oracle, SQL, and PySpark to support scalable transformation, querying, analytics, and downstream reporting workloads.
- Developed scalable, maintainable data processing workflows using SQL, Python, PySpark, Parquet-oriented patterns, schema discipline, and partition-aware design.
- Optimized SQL queries, Oracle schemas, Redshift-backed workloads, indexes, views, partitioning patterns, and reporting data models for performance, maintainability, and cost-aware processing.
- Built data quality, reconciliation, and validation practices across BMC TrueSight, CA Wily, AppDynamics, Oracle, and AWS-backed reporting layers to improve completeness, consistency, and trust in pipeline outputs.
- Troubleshoot production data and reporting issues by tracing source feeds, transformation logic, SQL behavior, failed outputs, latency symptoms, and downstream reporting discrepancies.
- Created technical documentation, metric definitions, runbooks, transformation notes, and support procedures to improve knowledge sharing and operational consistency.
- Partnered with infrastructure, application, analytics, and business stakeholders to establish repeatable delivery practices, mentor teammates, and align technical solutions with business needs.

Performance Engineer at G6 Hospitality LLC (2017-03 – 2017-11)

- Developed analytics and reporting outputs from Dynatrace AppMon and Gomez Synthetic Monitoring data for Brand.com and critical hospitality systems.
- Led the FAST project, mining real-user and synthetic performance metrics to identify optimization opportunities and communicate findings to engineering teams.

- Built dashboards and operational reporting views that converted raw monitoring data into actionable performance and reliability insights.
- Supported production system reliability by correlating application telemetry, transaction behavior, and infrastructure performance trends.

SME for CA APM (Senior Consultant) at CA Technologies / TIAA-CREF (2011 – 2016)

- Managed enterprise telemetry environments with 50+ Enterprise Managers and 4,000–6,000 instrumented agents, supporting monitoring, operational analytics, dashboards, and production reliability.
- Built Perl and Korn shell automation for extracting, transforming, and distributing performance data, replacing manual exports with repeatable reporting workflows.
- Established reusable dashboard, alert, SLA threshold, and documentation standards adopted across client environments.
- Mentored technical teams and partnered with J2EE, WebLogic, infrastructure, database, and operations groups to analyze data, troubleshoot bottlenecks, and resolve production issues.

Performance Test Engineer at AT&T (2010-08 – 2011-07)

- Analyzed enterprise telecom applications under load to identify throughput limits, SQL/JDBC bottlenecks, CPU, memory, thread, and garbage-collection constraints.
- Configured JMX Monitoring, Wily Introscope, and thread-dump automation to improve runtime visibility during performance and regression testing.
- Documented baseline metrics, test findings, regression thresholds, and release-readiness notes used by engineering teams for production decisions.

Senior Systems & Data Migration Engineer at Sabre (2008-05 – 2010)

- Led migration of a high-throughput shopping engine from 200+ MySQL nodes to a 6-node Oracle RAC cluster, preserving transaction performance and data integrity.
- Designed and optimized SQL and Oracle access patterns for high-volume workloads while reducing physical hardware footprint by 95%.
- Built C++/OCI/OCI validation and regression testing utilities to verify correctness across millions of migrated transaction records.
- Automated schema conversion, data validation, and phased cutover support for a mission-critical enterprise migration.

Architect / Developer (IRS CADE Project) at Computer Science Corporation (CSC) (2007-10 – 2008-05)

- Designed UML class and sequence diagrams and developed modules for IRS CADE mainframe modernization.
- Built CICS, MQ Series, XML messaging interfaces, and DB2-backed components in a government modernization environment.
- Supported technical design documentation and integration patterns for structured data exchange across modernization components.

Developer / Support Engineer (Billing, Interfaces, CSM/PRMS) at Corpus Inc. / Sprint (2001 – 2007)

- Developed and supported high-availability C++/Pro*C/PL/SQL billing interfaces and reporting processes for telecom-scale transaction systems.

- Worked with XML, SQL, Oracle, PL/SQL, messaging, and integration interfaces supporting billing, customer-management, and operational reporting workflows.
- Improved database performance by 10x through SQL, sequence, and process optimization while reducing memory usage by 75%.
- Built automation scripts in Korn shell, Perl, Awk, Syncsort, and SQL to support data extraction, transformation, reporting, deployments, and operational housekeeping.

Education

Post-Graduate Diploma in Computer Engineering Technology / Computer Science

Humber College, Toronto, Canada

Bachelor of Science in Civil Engineering

Zagazig University, Egypt

Skills

Languages: Python, SQL, PySpark, PL/SQL, XML, JSON, Perl, Korn Shell, C++, Java, C#
secondary/legacy exposure

Flagship Projects

Serverless Lakehouse Platform (AWS)

Designed a scalable AWS data platform pattern using S3, Glue, Athena, Redshift, PySpark, and Parquet for raw ingestion, transformation, curated analytics datasets, query optimization, reporting access, and validation checkpoints.

Technologies: AWS S3, AWS Glue, Athena, Redshift, PySpark, Parquet, SQL

HorizonScale — AI Capacity Forecasting Engine

Built a forecasting-oriented analytics workflow using Python, PySpark, Prophet, scikit-learn, and Streamlit to process infrastructure telemetry, prepare model-ready datasets, validate pipeline outputs, and deliver capacity insights.

Technologies: Python, PySpark, SQL, Prophet, scikit-learn, Streamlit, Data Validation

AI-Powered Job Search Pipeline

Built an AI-assisted Python automation pipeline with semantic duplicate detection, LLM-based scoring, JSON artifact generation, quality gates, document rendering, and application tracking using Claude Code, Grok/xAI APIs, FAISS, and sentence-transformers.

Technologies: Python, Claude Code, Grok / xAI API, FAISS, Sentence Transformers, JSON, Quality Gates